



AUTOMATIC WASTE SEGREGATION

Present by:

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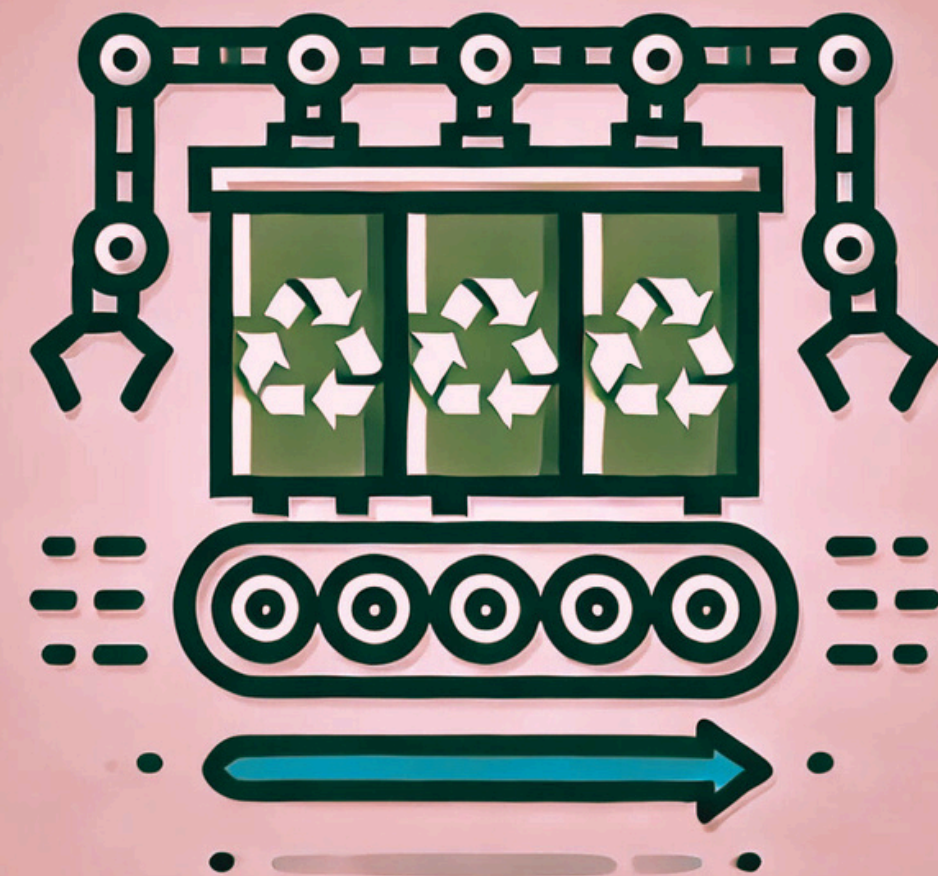
Abdallah Mahamat Abacar Maeyd (2117777)





WHAT IS AUTOMATIC WASTE SEGREGATION?

Automated waste disposal uses AI, robotics, and machine learning to sort waste faster and more efficiently, saving time and improving environmental control.



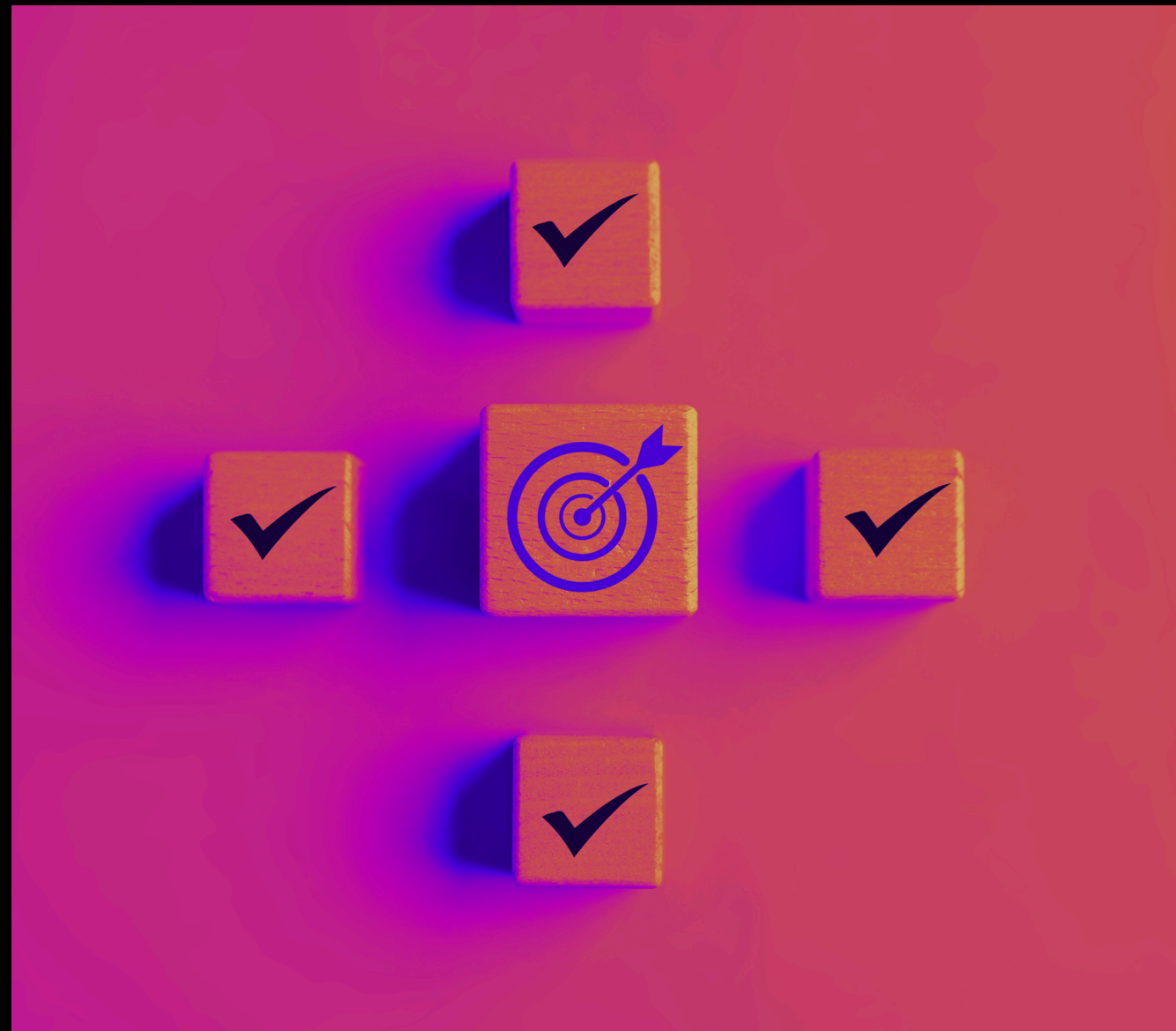


Problem Statement

Public places need an automated waste segregation system to reduce pollution, improve recycling, and lower waste management costs by accurately sorting materials while minimizing labor and errors.



Objectives



01

To design a machine that can segregate recycled waste (Paper, glass, plastic, and metal) in minimizing the manual efforts of sorting.

02

To evaluate performance of the machine



Customer survey



This survey focuses on target market, waste type, segregation frequency, segregation challenges, difficult waste types, automation importance, accuracy expectations, space availability, investment willingness, user intervention, real-time feedback, and open feedback

Some of the questions

Who is answering this survey?

What type of waste do you primarily generate?

How often do you deal with waste segregation or disposal?

If you do segregate waste, what challenges do you face?

Which types of waste are the most challenging for you to segregate?

How important is it for you to have an automated system that can handle multiple types of waste simultaneously?

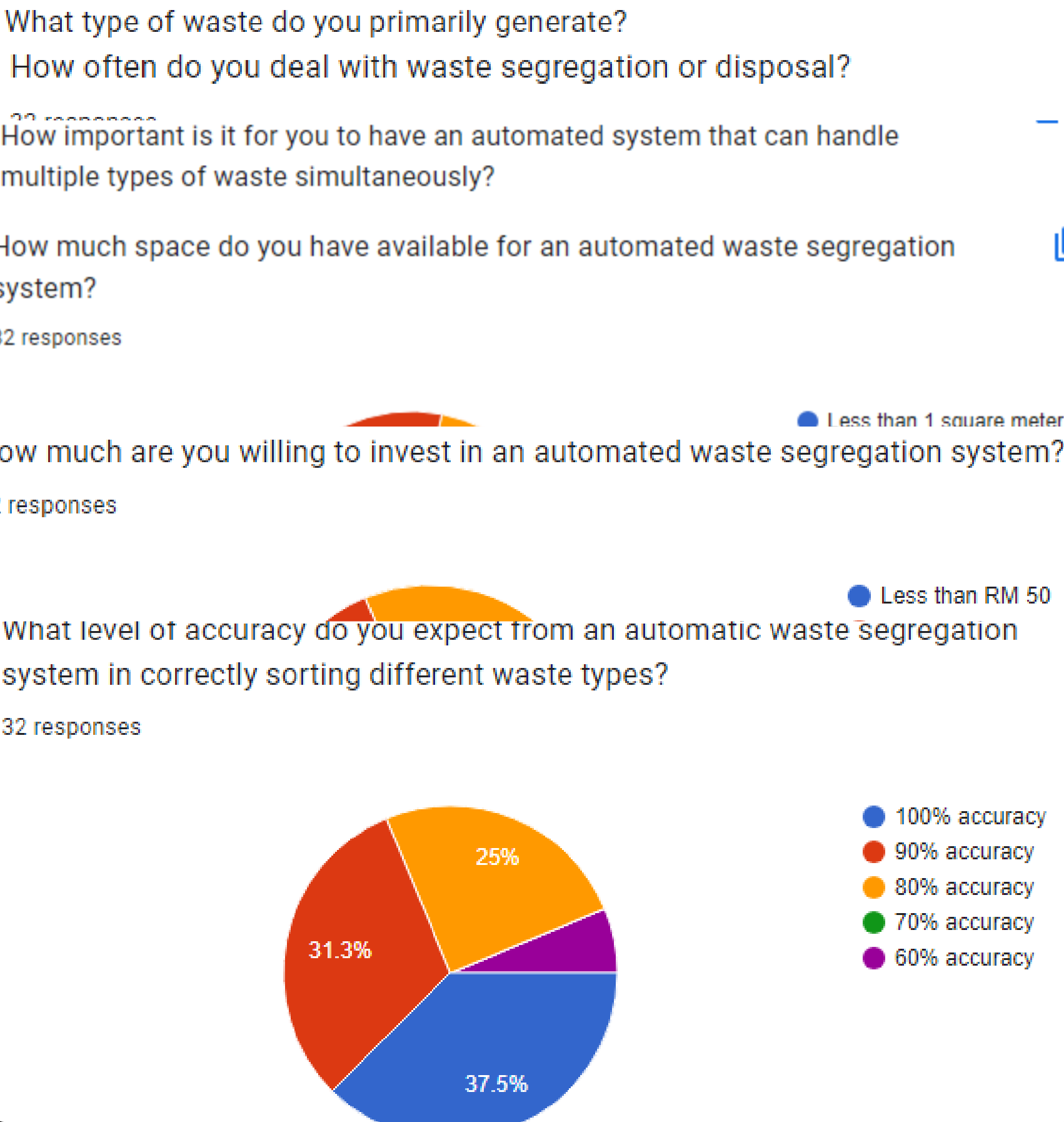
What level of accuracy do you expect from an automatic waste segregation system in correctly sorting different waste types?

How much space do you have available for an automated waste segregation system?

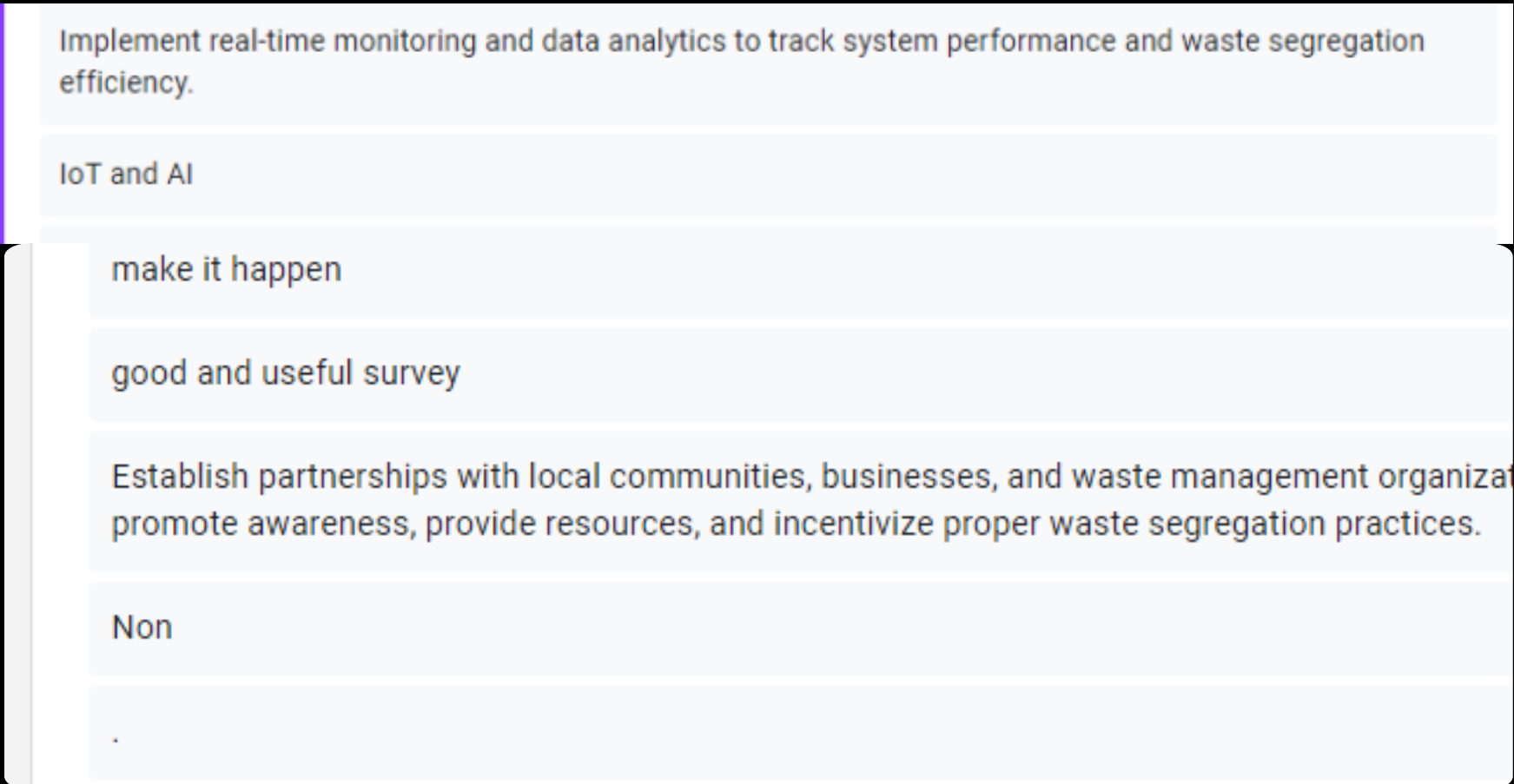
How much are you willing to invest in an automated waste segregation system?



Customer survey

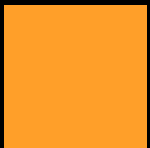


RECOMMENDATIONS





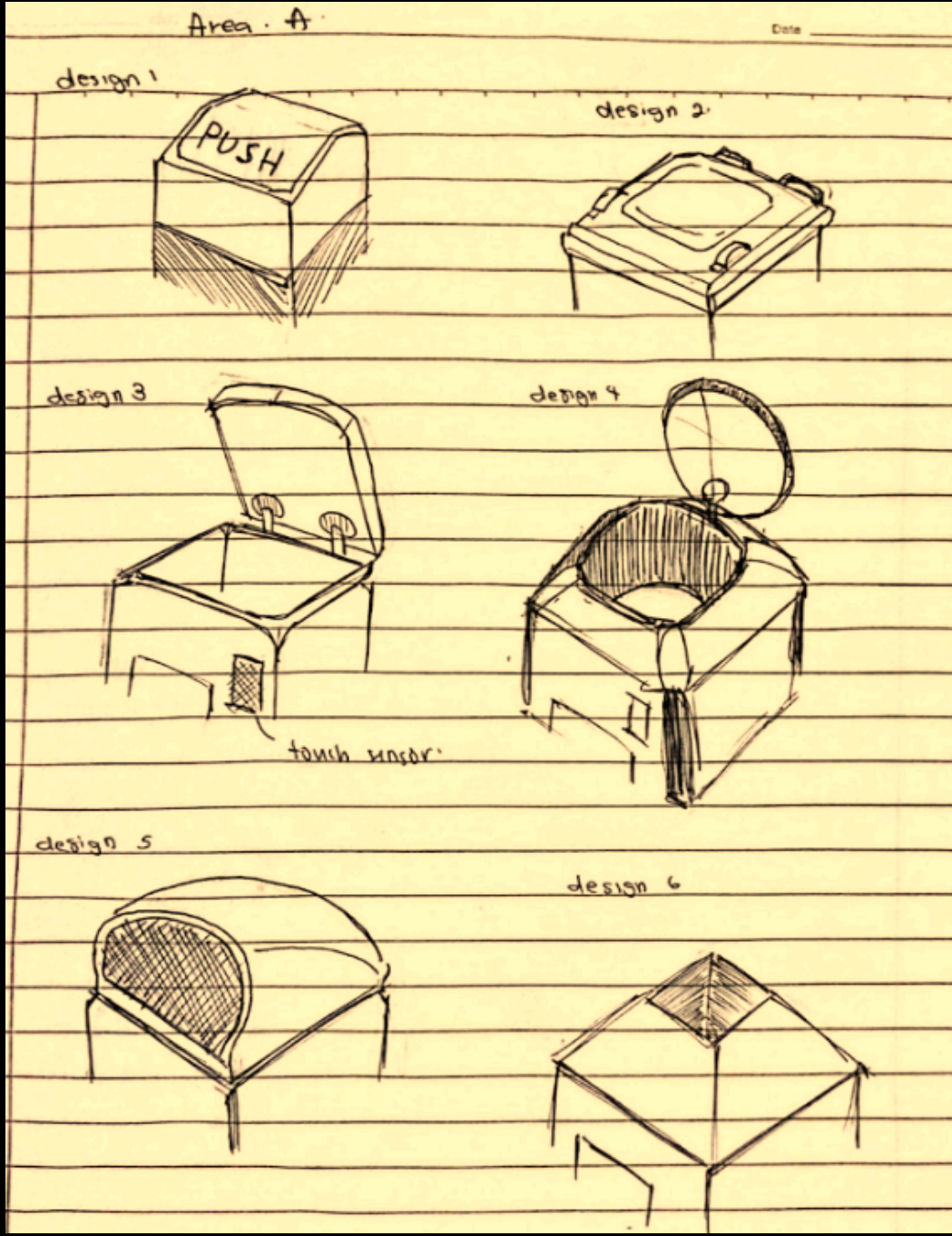
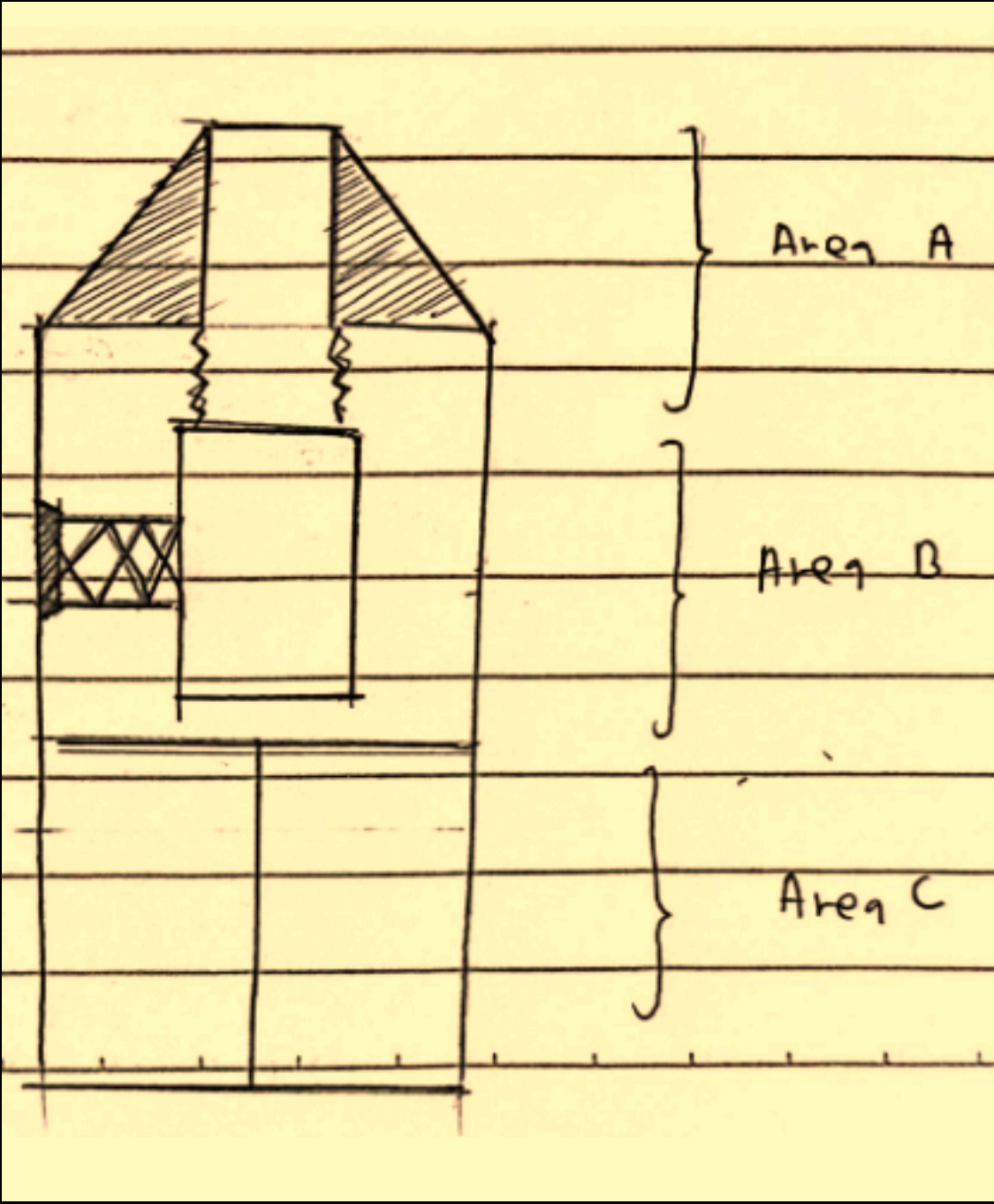
Design of The System



Concept Design

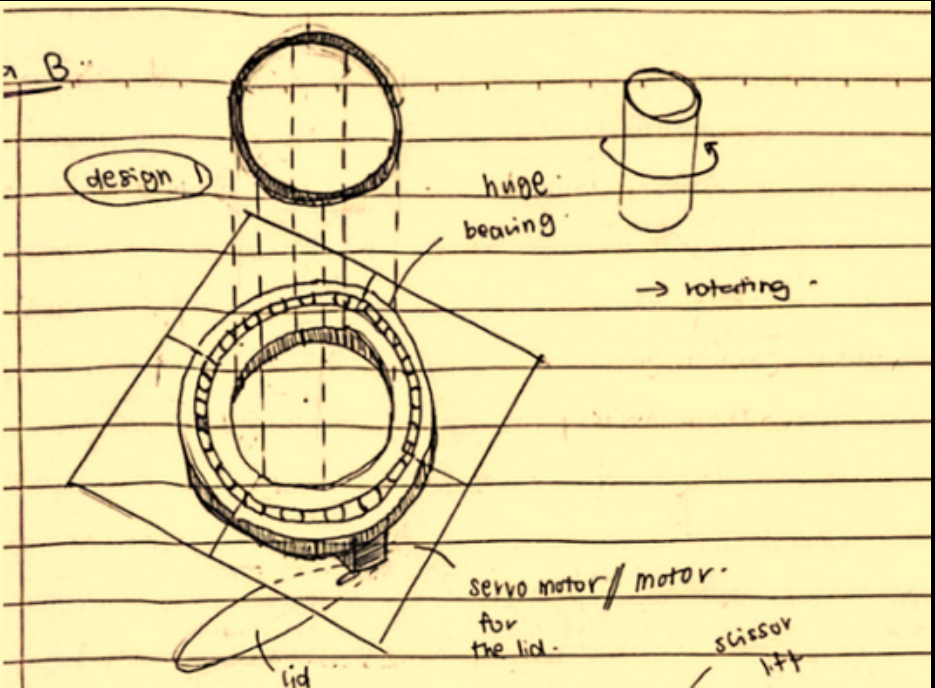


Area A (the head)

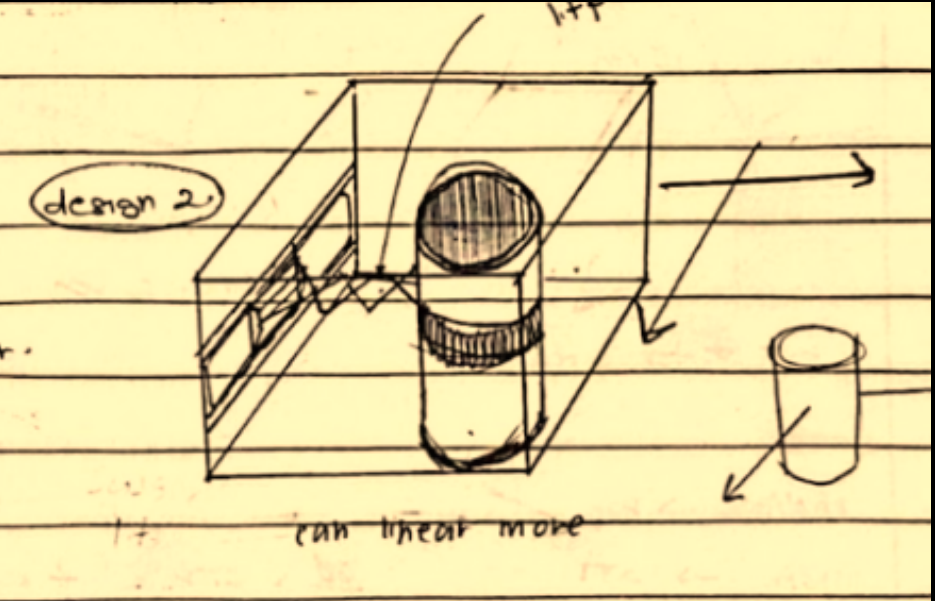


Area B (the body)

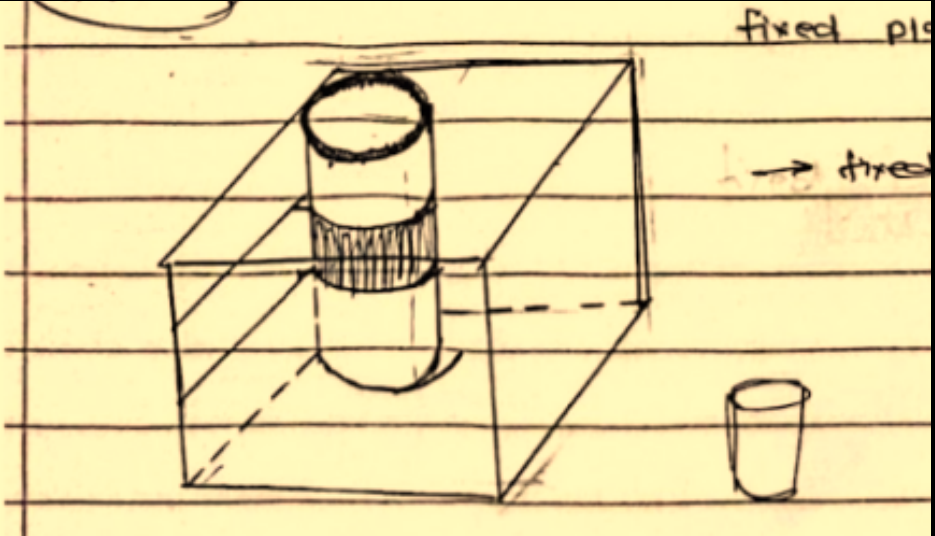
1



2



3

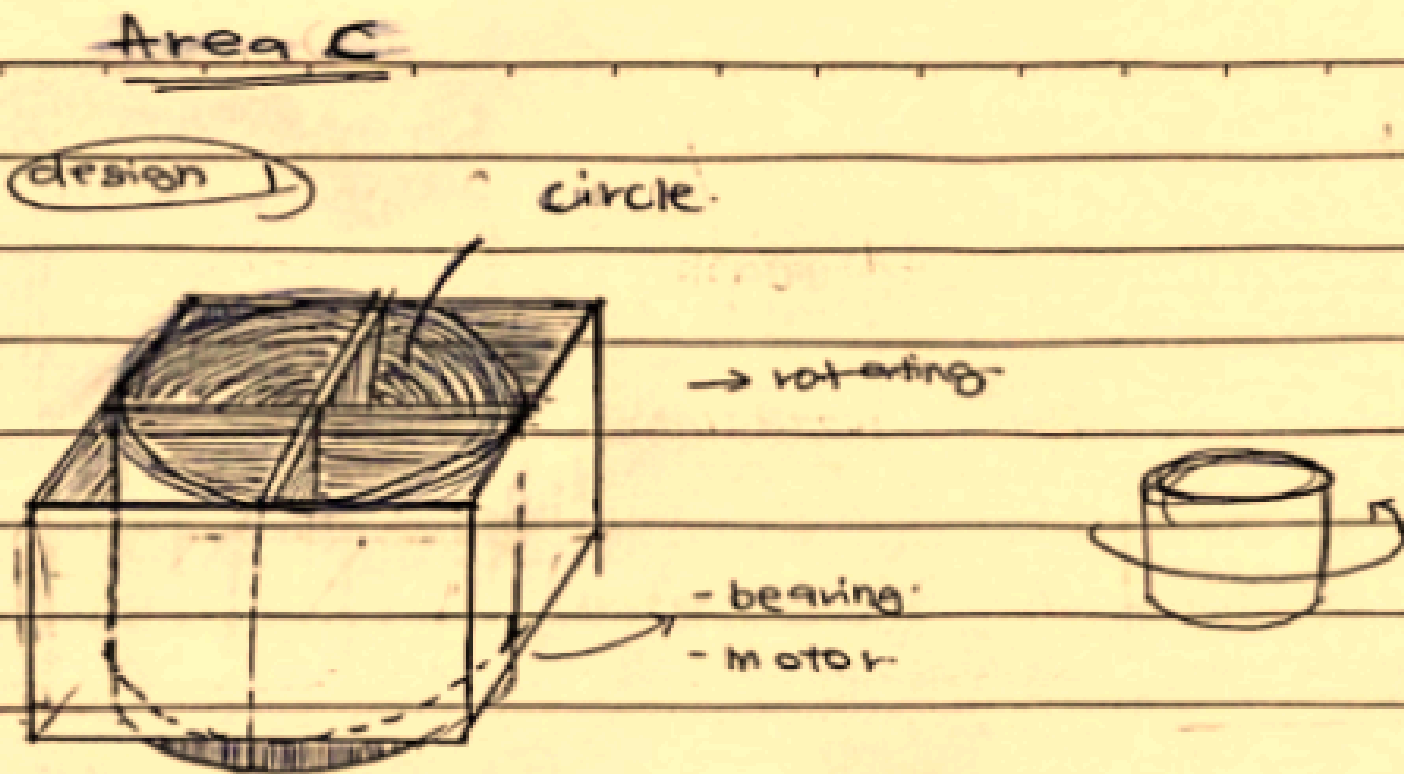




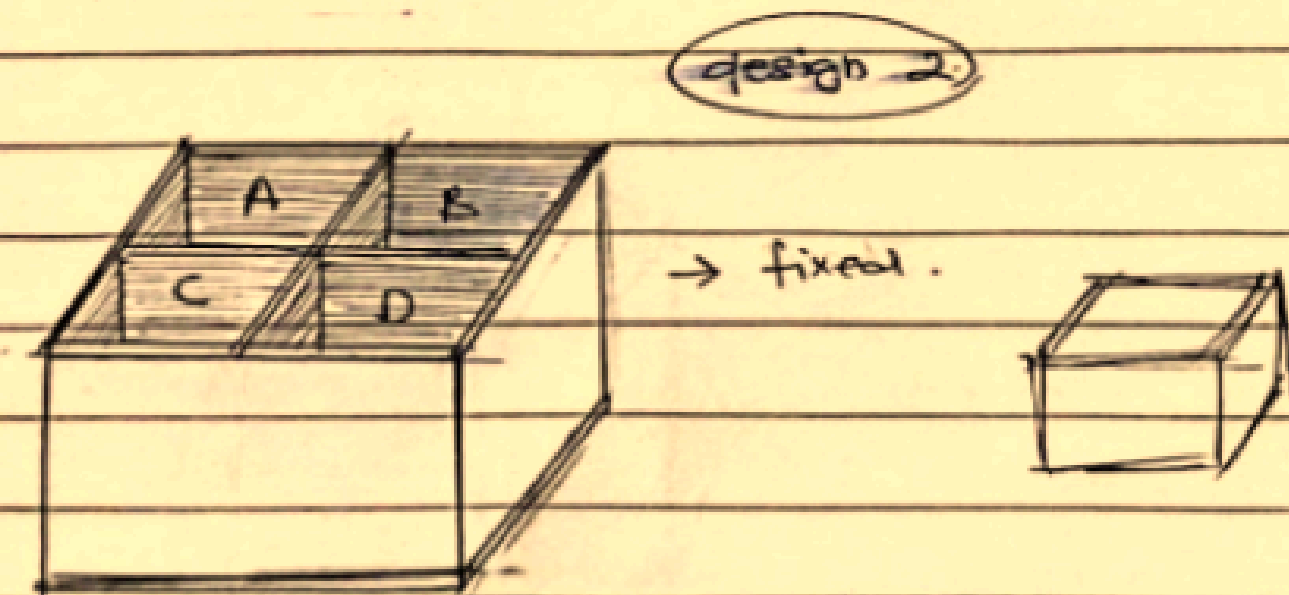
Design of The System

Area C (the body)

1



2



- **First Combination (Design 1) :**

Area B: Design 2 (linear movement mechanism similar to a 3D printer)

Area C: Design 2 (fixed compartments)

- **Second Combination (Design 2) :**

Area B: Design 3 (fixed cylinder)

Area C: Design 1 (rotating compartments with bearing and motor)

- **Third Combination (Design 3) :**

Area B: Design 1 (rotating mechanism with servo motor)

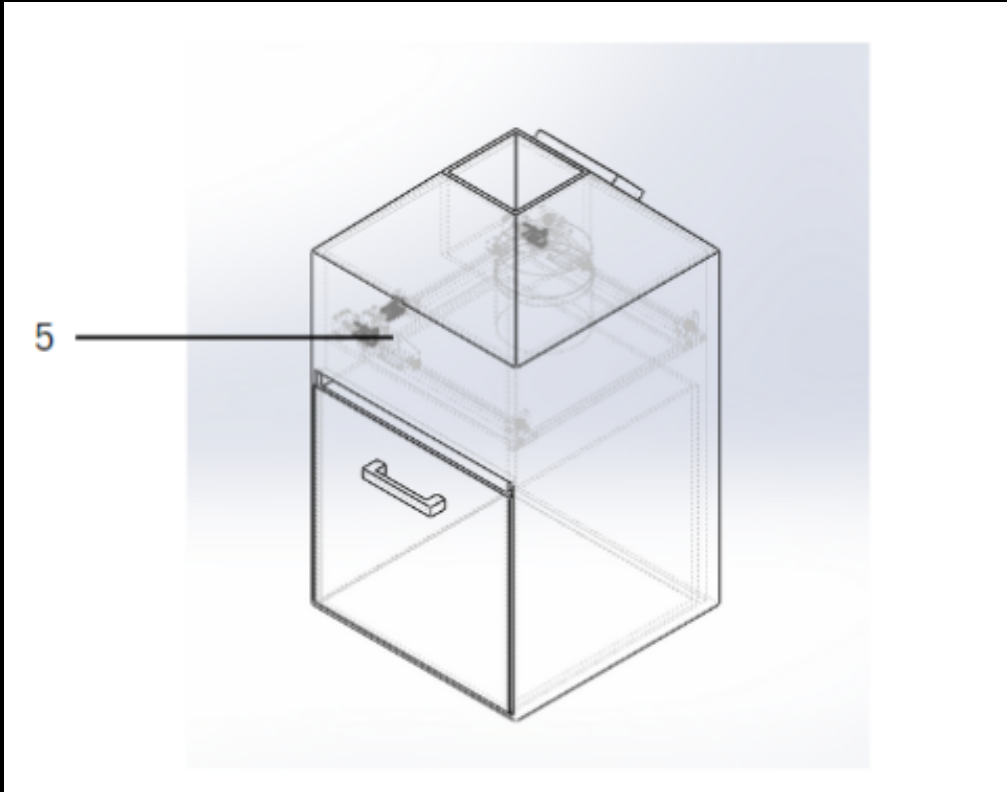
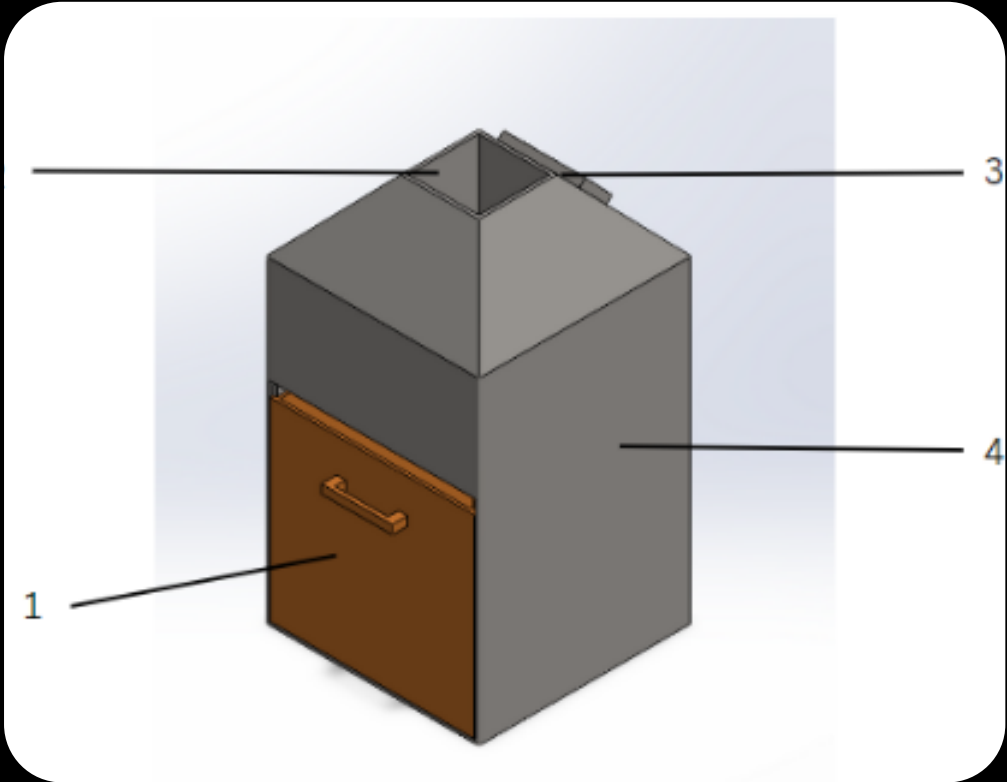
Area C: Design 2 (fixed compartments)

Our chosen design is 1 based on Pugh Chart method



FINALIZED PRODUCT

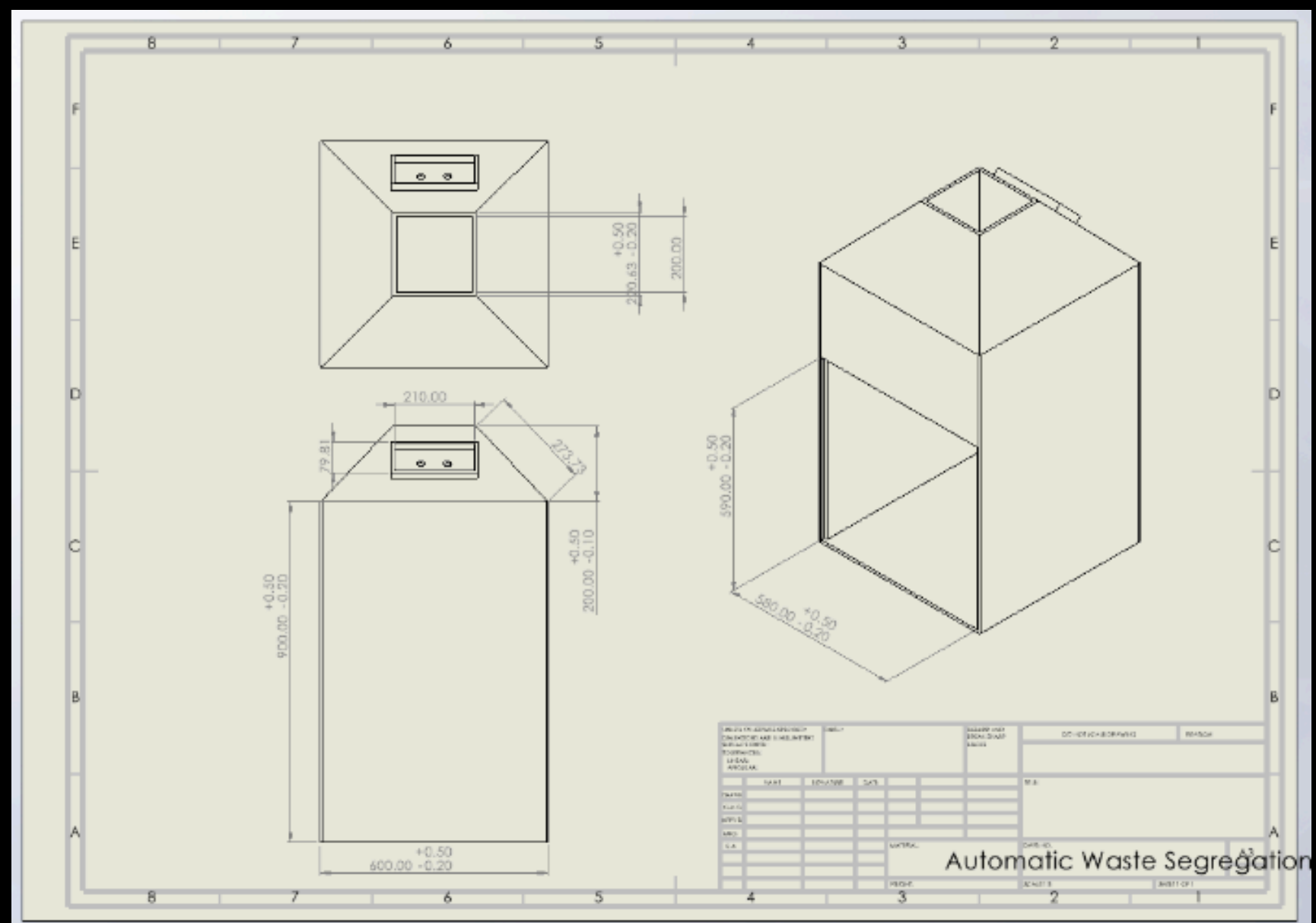
1	frame
2	Storage Box
3	Segregate Mechanism
4	Input Hopper
5	User Interface Section



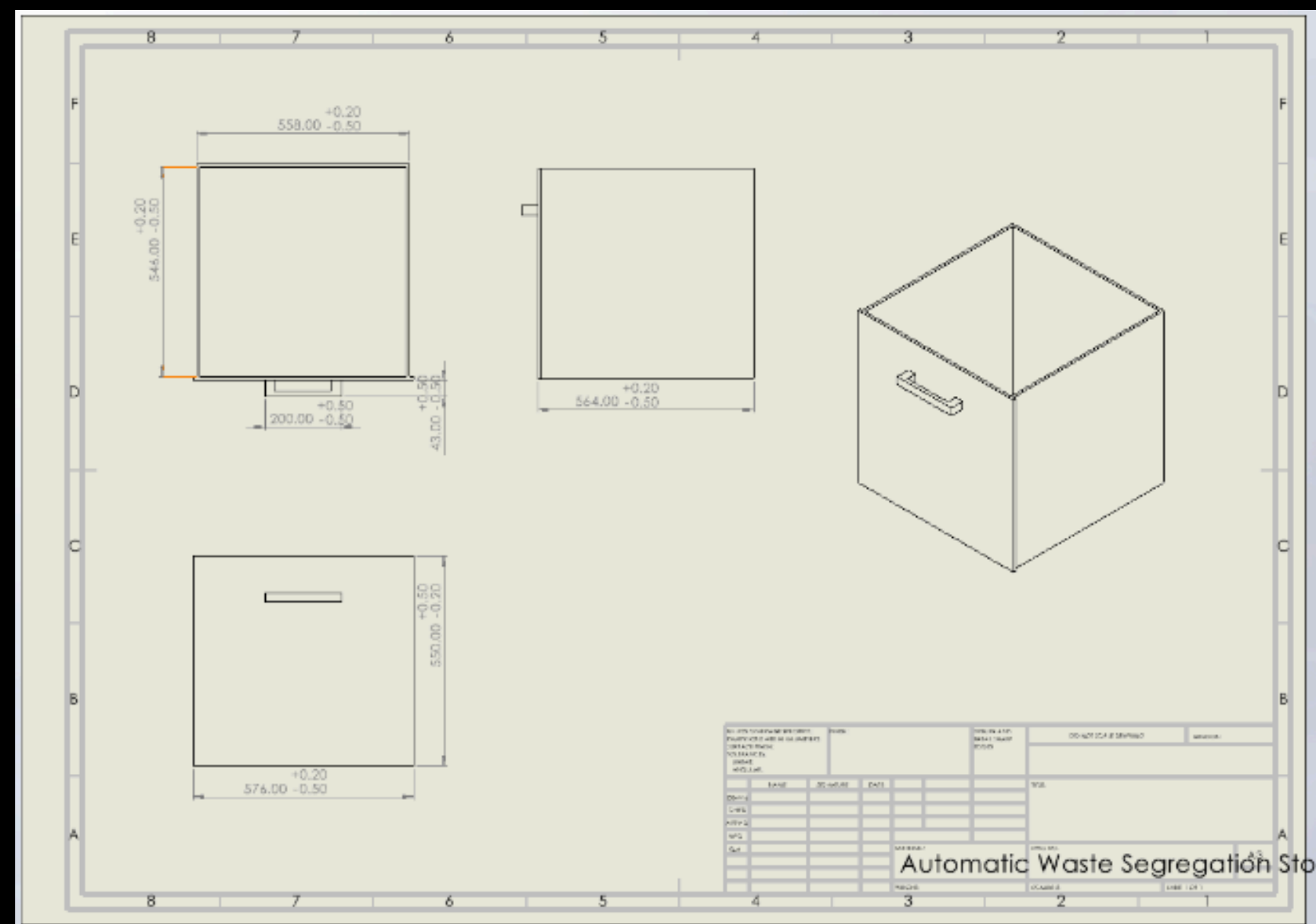


FINALIZED PRODUCT

MAIN FRAME



STORAGE BOX

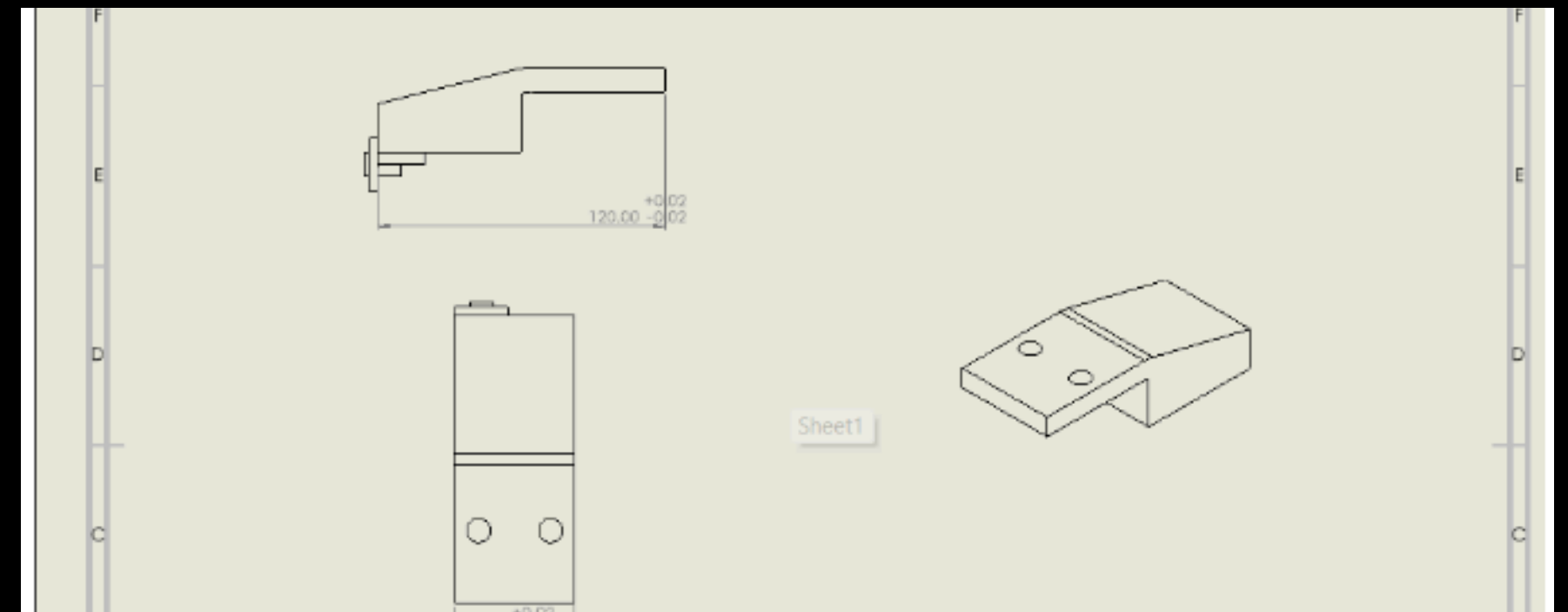
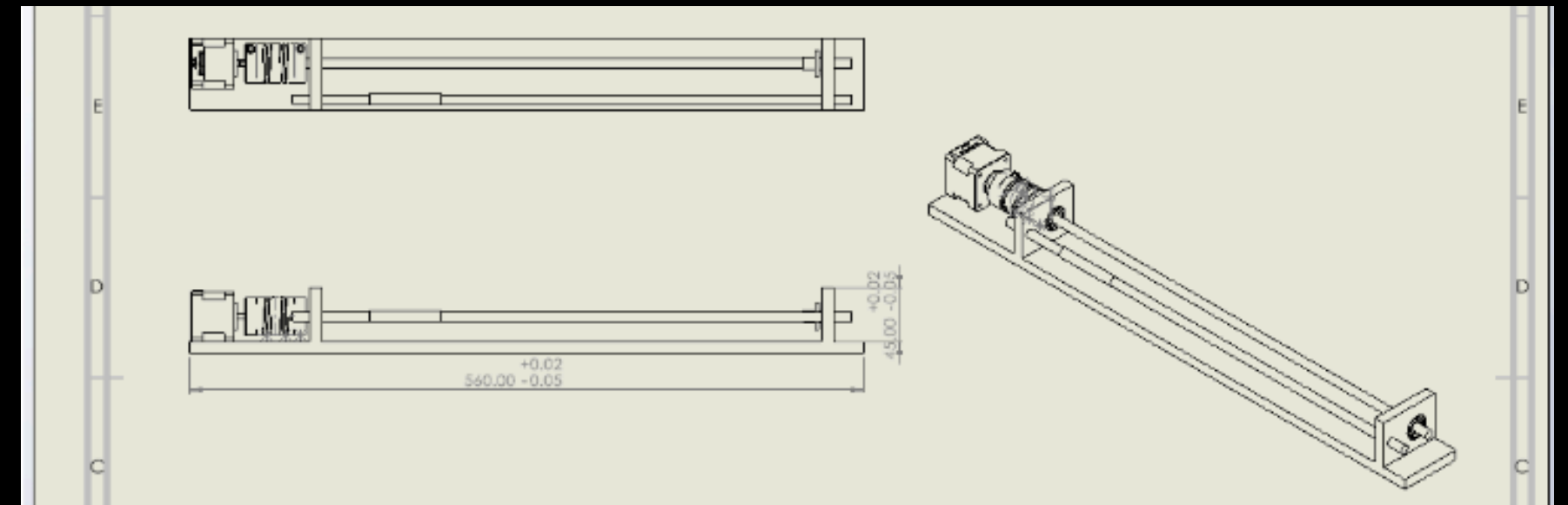
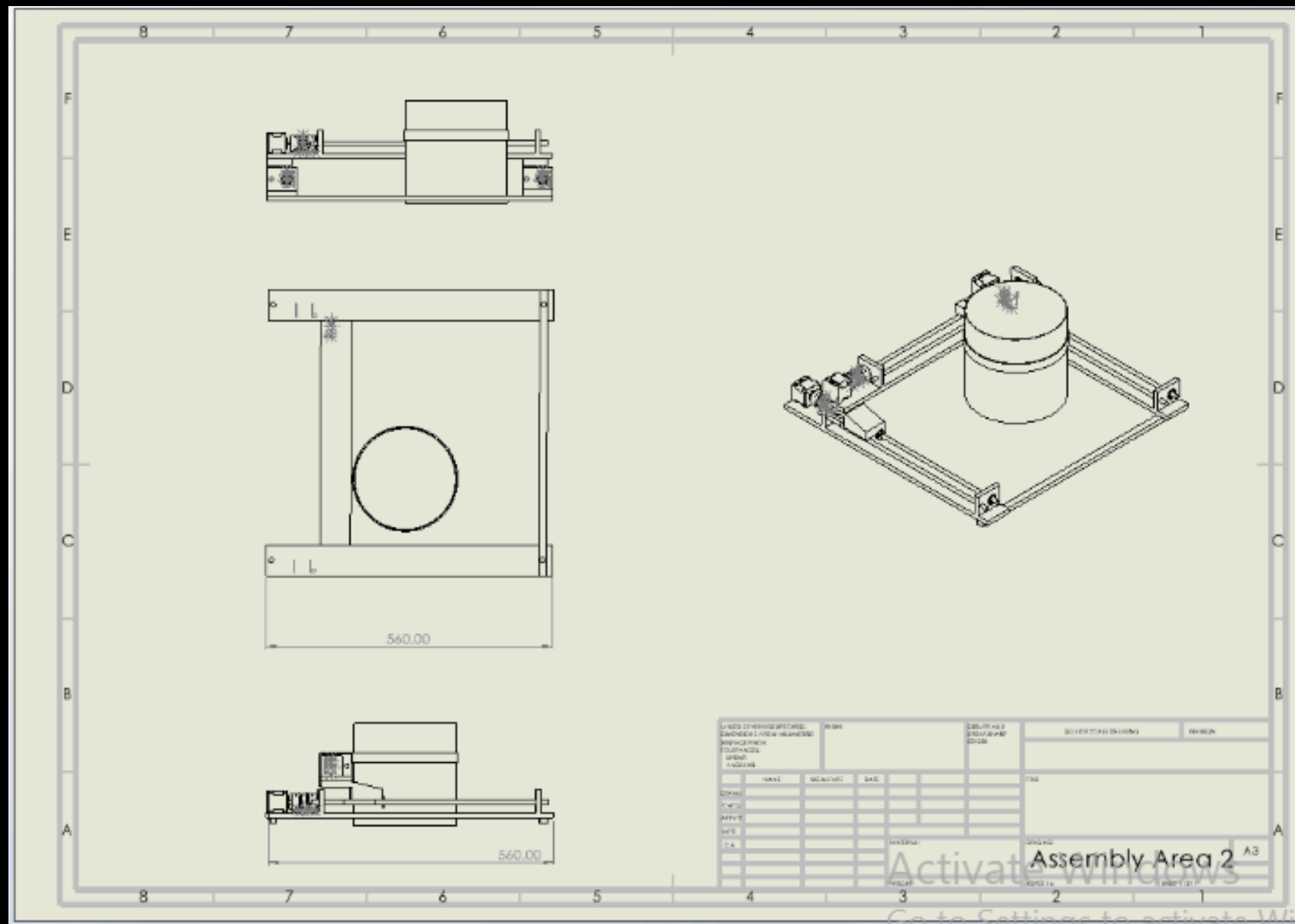




FINALIZED PRODUCT

SEGREGATE MECHANISM

STEPPER MOTOR PLATFORM





LIST ITEM

Microcontroller	● Model : Arduino Mega 2560 ● Processor : ATmega2560 ● Digital I/O Pins : 54 (15 PWM outputs) ● Analog Input Pins : 16 ● Operating Voltage : 5V ● Flash Memory : 256 KB ● SRAM : 8 KB
Motor driver	● Model : DRV 8825 Stepper Motor Driver ● Voltage : 8.2V to 45V ● Current : 1.5A per phase (2.2A peak) ● Microstepping : Up to 1/32 step ● Interface : Step and direction control
Bipolar Stepper Motor	● Model : NEMA 17 ● Step Angle: 1.8° per step ● Holding Torque: 45 N·cm (63.74 oz·in) ● Current: 1.5A per phase ● Voltage: 12V
Servo Motor	● Model : MG996R ● Operating Voltage: 4.8V - 7.2V ● Stall Torque: 9.4 kg·cm (4.8V), 11 kg·cm (6V) ● Speed: 0.19 sec/60° (4.8V), 0.14 sec/60° (6V) ● Rotation: 180°

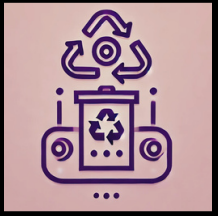
Led	● Model : 5mm LED ● Operating Voltage: 2V - 3V ● Current: 20mA ● Colours: Red, Green, Blue (RGB option available)
LCD	● Model : 16x2 LCD (HD44780) ● Operating Voltage: 5V ● Interface: Parallel ● Backlight: LED backlight ● Character Size: 5x8 dots
Wire	● Type: Black Red Wire ● Length: 20m ● Gauge: 22 AWG
Frame	● Material : Stainless Steel or Durable Plastic ● 600mm x 600mm x 1000mm (cuboid shape) ● Thickness : 20mm x 20mm ● The whole main body
Breadboard	● Size: 830 tie-points ● Dimensions: 16.5 x 5.4 x 0.85 cm ● Rows: 2 power rails, 63 rows ● Columns: 10
Ultrasonic sensor	● Model : HC-SR04 ● Operating Voltage: 5V ● Measuring Range: 2cm - 400cm ● Resolution: 0.3cm ● Angle: 15 degrees



LIST ITEM

Camera	● Model : Raspberry Pi Camera Module V2 ● Resolution: 8 Megapixels ● Video: 1080p at 30fps ● Interface: CSI connector
Power Adapter	● Input: 100-240V AC, 50/60Hz ● Output: 12V DC, 2A ● Connector: Barrel Jack
Lids	● Material: Stainless Steel or Durable Plastic ● Shape : Square ● Automatic with Touch Sensor
Flexible Shaft Coupling	● Model : Aluminum Flexible Shaft Coupling ● Outer Diameter (OD): 19mm ● Length (L): 25mm ● Material: Aluminium Alloy ● Flexibility: Can accommodate slight angular, parallel, and axial misalignments ● Torque Rating: Up to 2 Nm ● Usage: Connects motor shafts to lead screws or other shafts, reduces misalignment stress.
Round Wooden Stick	● Material : Hardwood ● Diameter: 8mm (matching the coupling and bearing specifications) ● Length: 500mm ● Finish: Sanded smooth, optionally treated with a clear finish or varnish for durability ● Usage: Can be used as a non-metallic, lightweight alternative for structural or mechanical components.

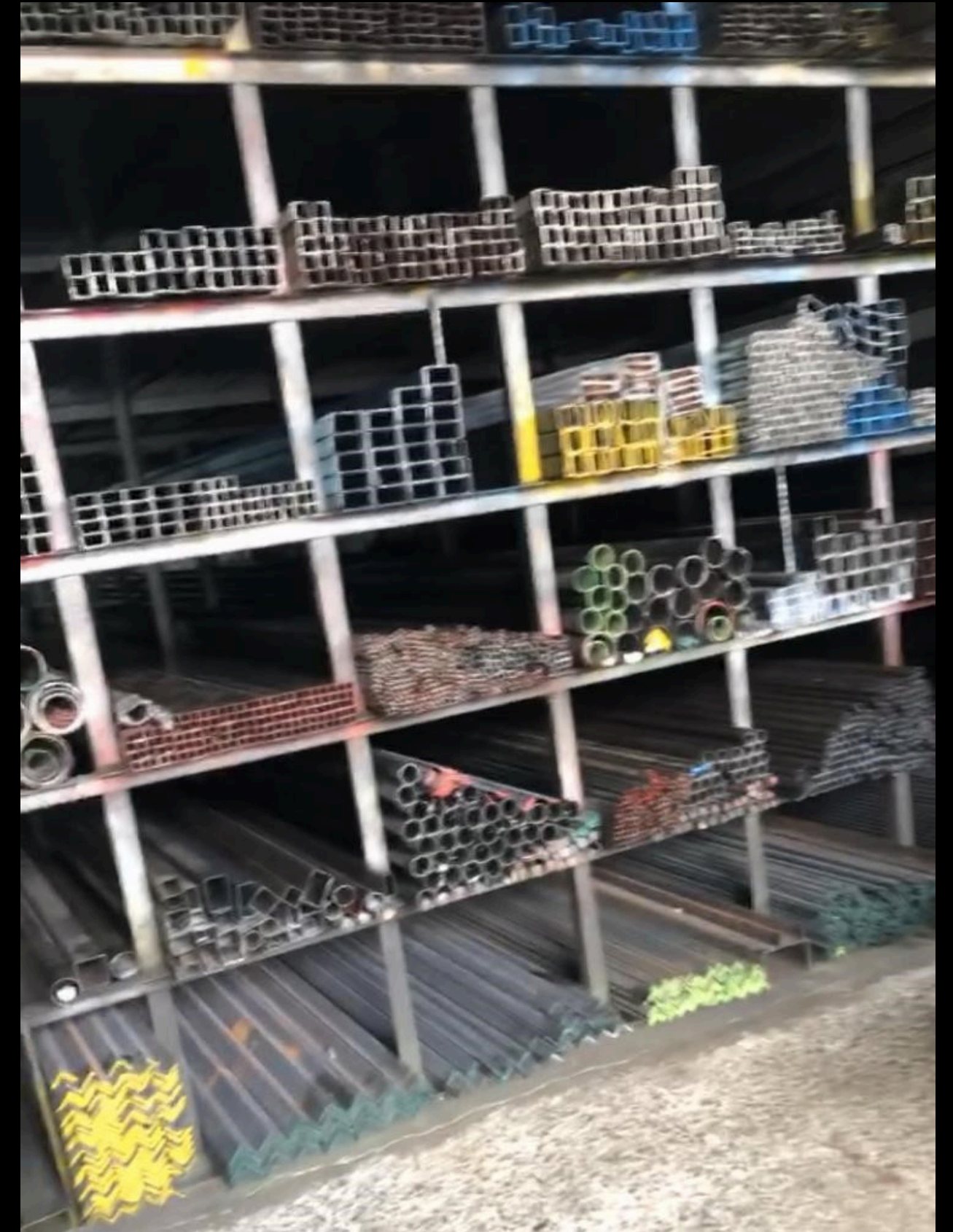
Battery Backup	● Model: Li-Ion Rechargeable Battery Pack ● Voltage: 12V ● Capacity: 2000mAh ● Output Current: 1A
Microcontroller shield	● Model : Arduino Mega Shield for stepper motor drivers (CNC Shield V3) ● Compatible with Arduino Mega 2560 ● Designed for easy connection of multiple stepper motor drivers, such as the DRV8825 ● Voltage Range: Typically 12-36V DC ● Jumpers to set microstepping resolution (up to 1/32 step with DRV8825) ● Supports up to 4 stepper motors (X, Y, Z, and an additional axis)
Bearing	● Model : 608ZZ ● Inner diameter : 8mm ● Outer Diameter (OD): 22mm ● Width: 7mm ● Material: Chrome Steel ● Shield: Metal Shields (ZZ) or Rubber Seals (RS) ● Load Rating: Dynamic: 3.6 kN, Static: 1.55 kN ● Max Speed: 34,000 RPM (for ZZ), lower for RS due to sealing friction ● Usage: Suitable for high-speed, low-noise applications; commonly used in linear motion systems.



system progression

1. Frame

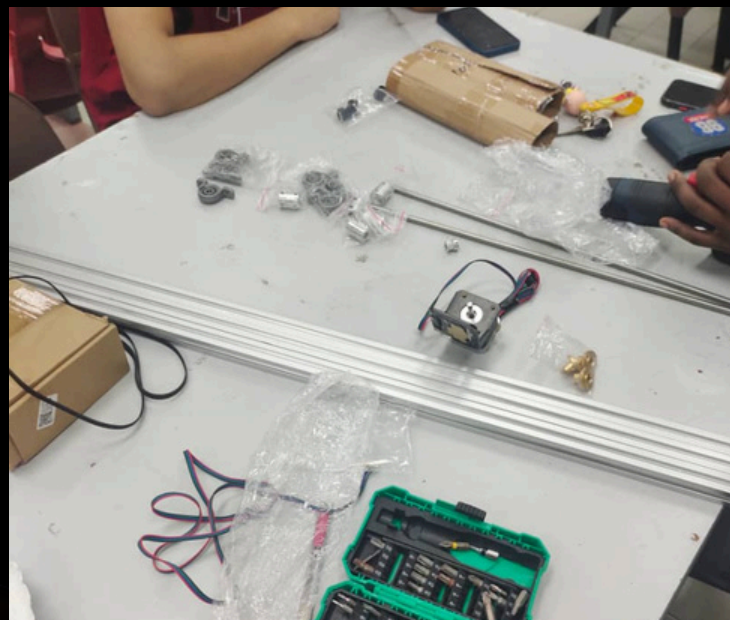
- Material: Stainless Steel
- 600mm x 600mm x 1000mm
(cuboid shape)
- Thickness : 30mm x 30mm
- The whole main body





system progression

1. Measure Dimension



2. Cutting & Welding Metal Components



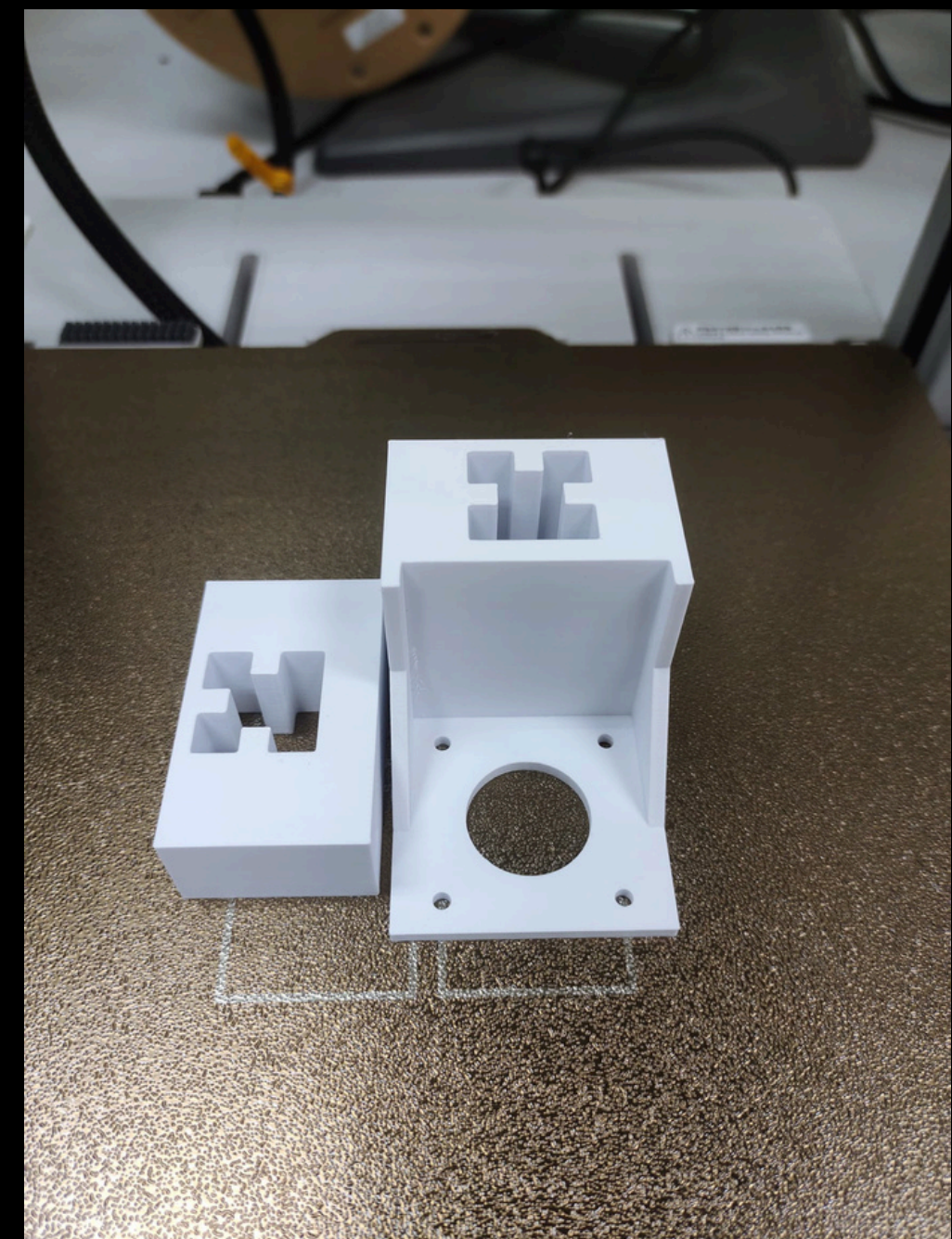
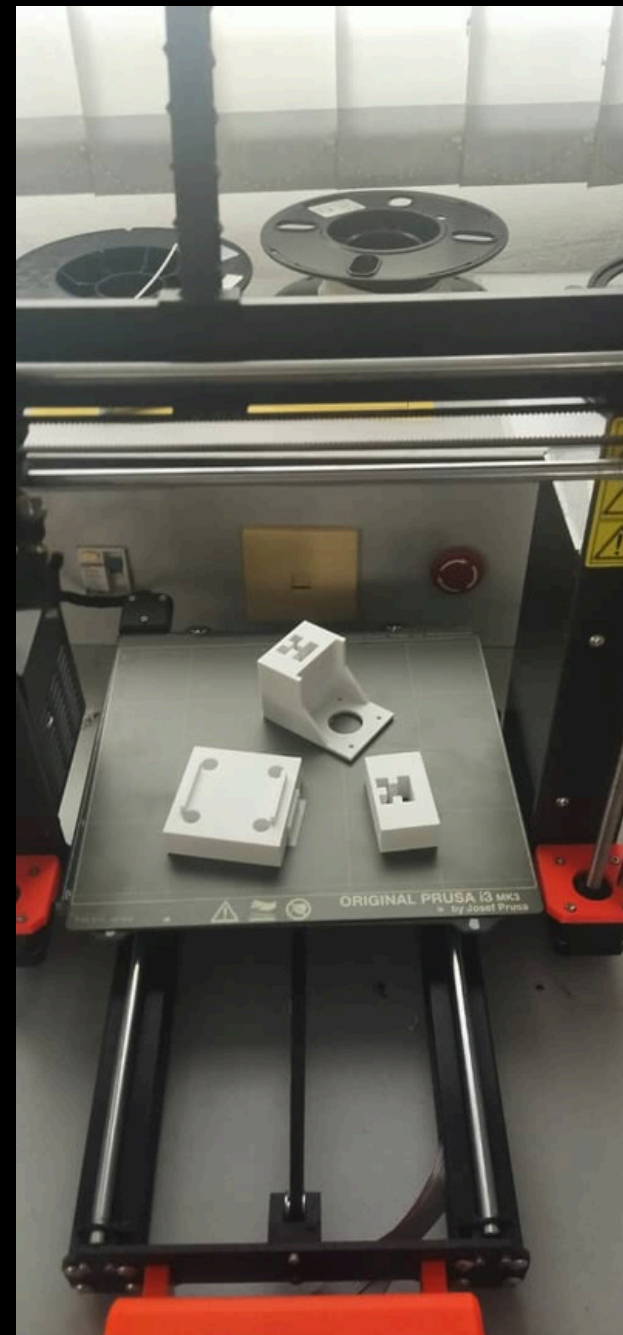
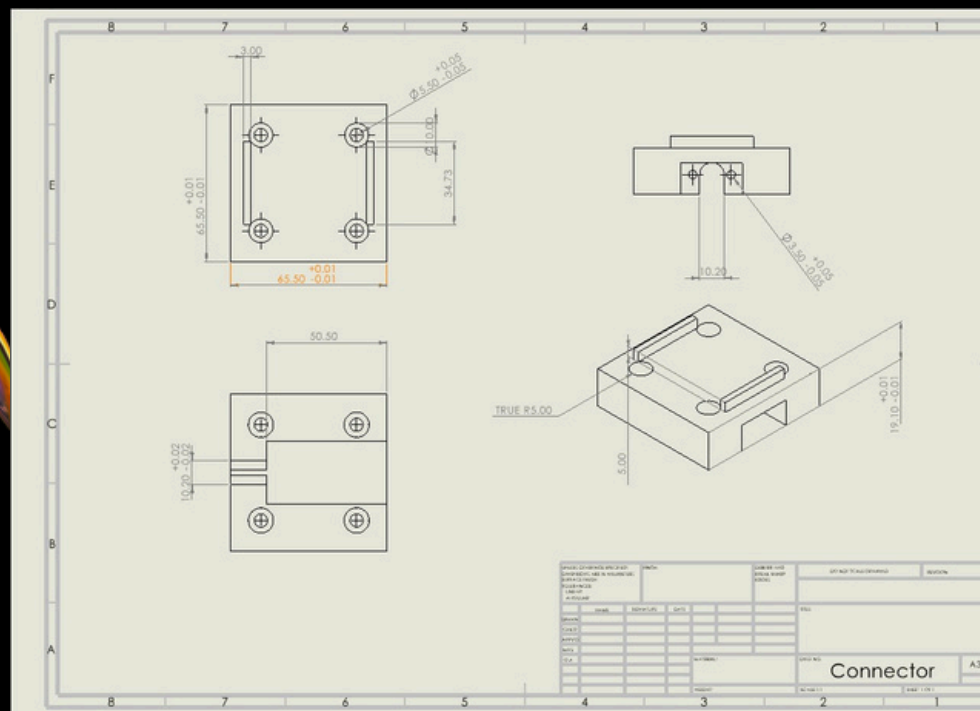
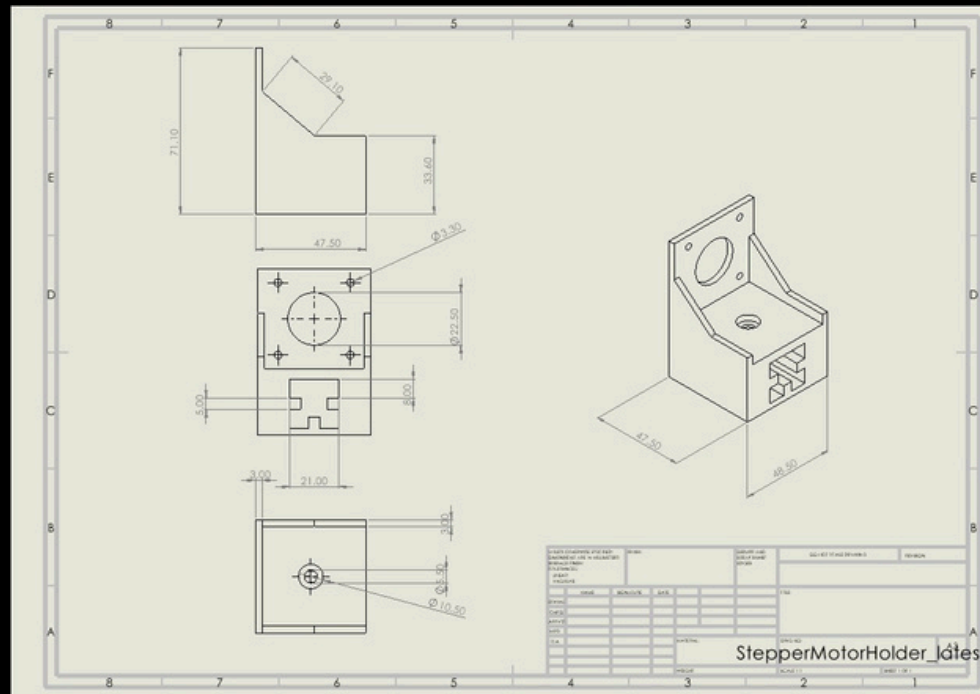


system progression

2. 3d printing Process

- supportings

- Final design

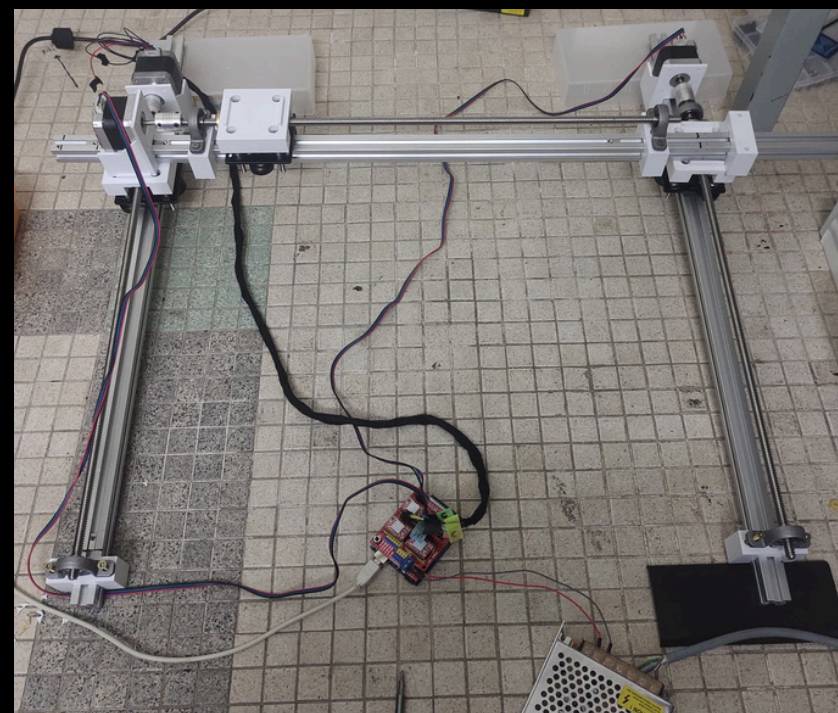




system progression

3. Segregation Mechanism

1. Fabrication Process & Assembly Process



2. Testing the Gantry system Process





Calculation

1. Load and Torque Calculation

Key Considerations:

1. Motor Specifications:

- Torque rating of the NEMA 17 stepper motors (typically 0.4–0.5 Nm for standard models at rated current).
- Current and voltage supplied by the driver.

2. Mechanical Characteristics:

- Lead screw pitch or belt pitch (linear movement per motor revolution).
- Friction and efficiency of the guide rails and transmission system.

3. System Configuration:

- Total load supported by Motor C (including its own weight, load, and frame).
- Torque distribution on Motors A and B if they share the load.

4. Load Movement Steps:

- Total steps to cover 5000 steps (as specified).



Calculation

Maximum force

- Using the relationship between torque and linear force on the leadscrew.
- The theoretical maximum force is applied by each motor before friction and efficiency losses.

$$F = \frac{2\pi \cdot T}{P}$$

F : linear force (N)
 T : motor torque (Nm)
 P : lead screw pitch mm/rev

$T = 0.4 - 0.5 \text{ Nm}$
 $F = ?$
 P : Assume a lead screw pitch 2mm/rev
 $P = 2 \text{ mm}$

So,

$$F = \frac{2\pi \cdot 0.5}{0.002} = 1256 \text{ N}$$

By using 80% efficiency (including frictions)

$$F_{\text{actual}} = 1256 \times 0.8 = 1004.8 \text{ N}$$



Calculation

Maximum mass

- If Motor C carries a smaller load (e.g., less than 102.4 kg), the total capacity increases slightly.

For maximum load (mass)

Motor C will lift the loads directly.

$$F = m_{\text{max},c} g$$

$$m_{\text{max},c} = \frac{F}{g}$$

$$g = 9.81 \text{ m/s}^2$$

$$F = 1004.8 \text{ N}$$

$$\text{So, } m_c = \frac{1004.8}{9.81} = 102.4 \text{ kg}$$



Calculation

distance calculation

- Steps for a Require Travelling
- assume (5000) steps

NO Date

Steps to travel a certine distance. assume (5000)

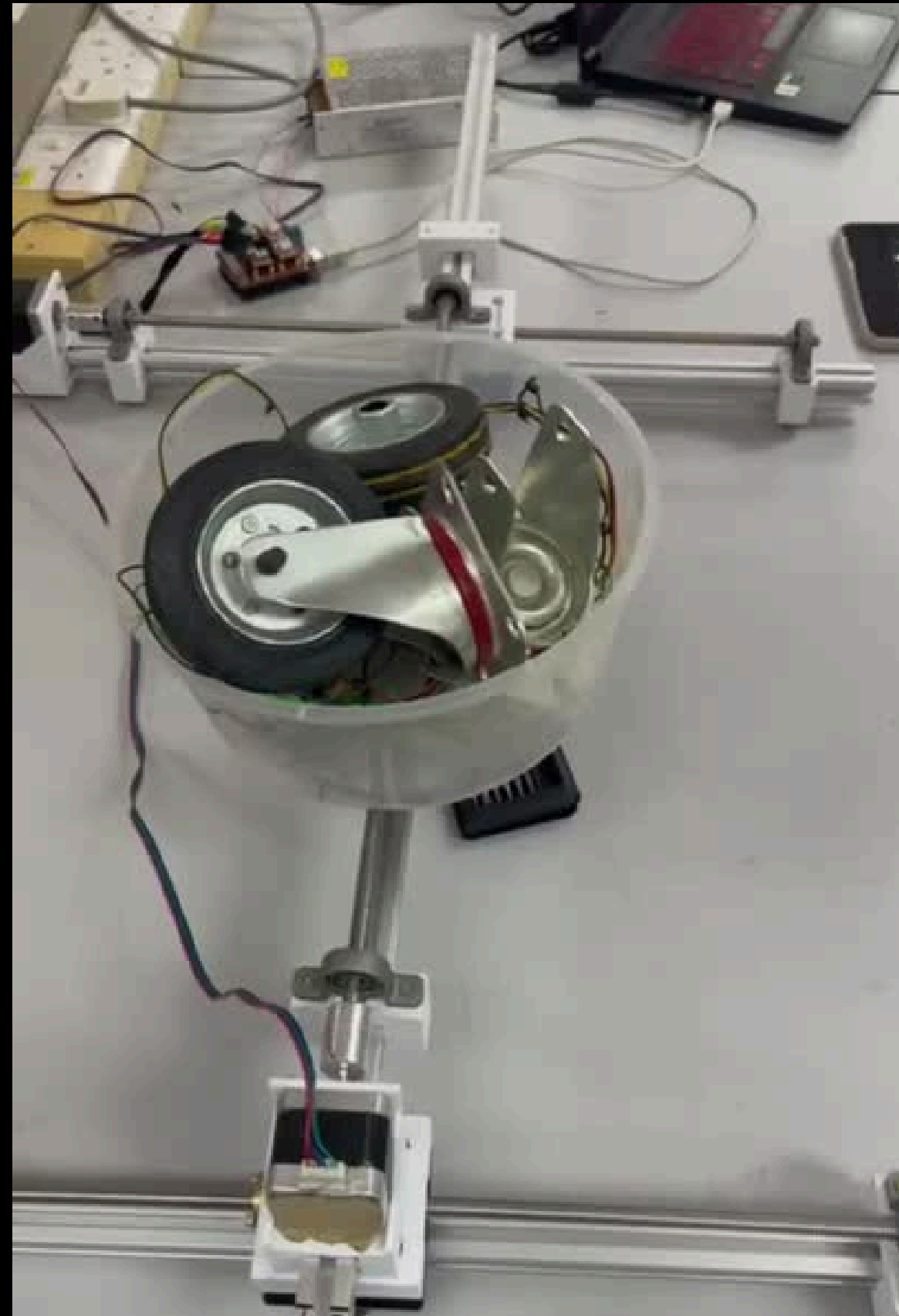
* Lead screw pitch: $0.002 \text{ m/rev} = P$
* steps of the motor 200/rev

$$D = \frac{5000}{200} \cdot P$$
$$D = \frac{5000}{200} \cdot 0.002 = 5 \text{ cm}$$



Calculation

testing
6kg of
mass



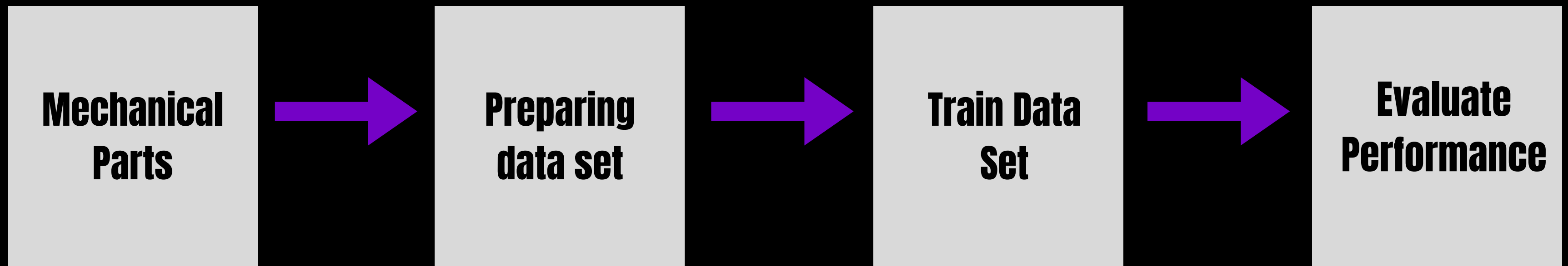


Machine Learning

Image Classification

- Using ML.NET in Visual Studio 2022 for Image Classification

FLOW





Machine Learning

1. Insert Dataset

Data Preview

Total images 1788. Showing 8/516.

TestingDataset

TrainingDataset

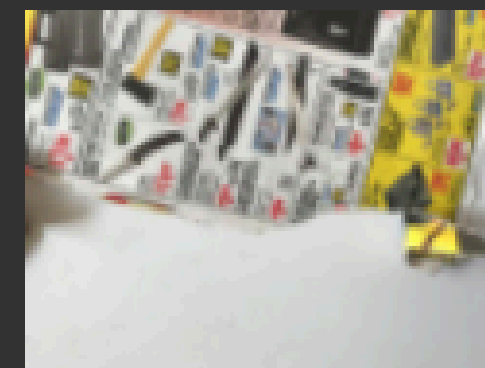
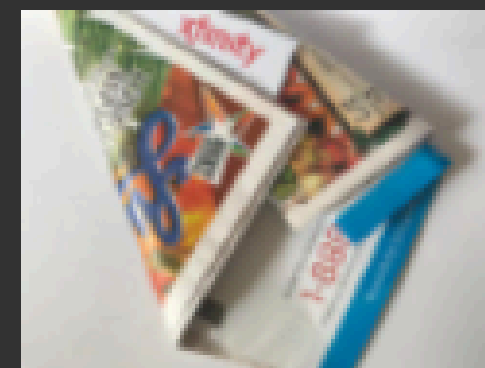
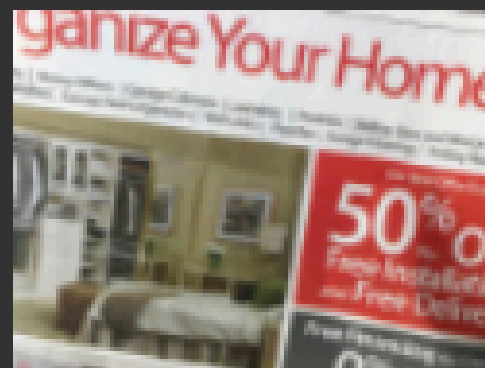
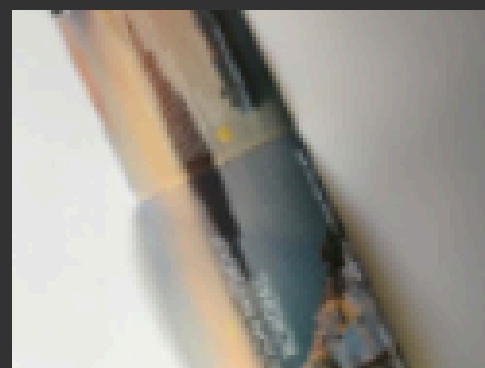
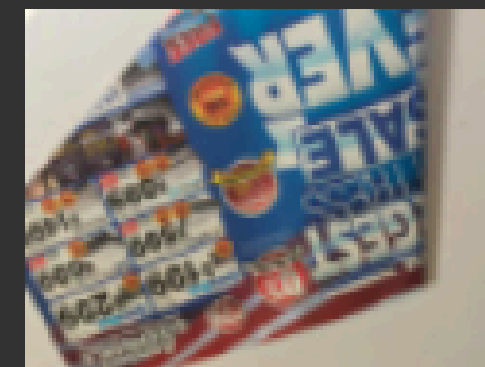
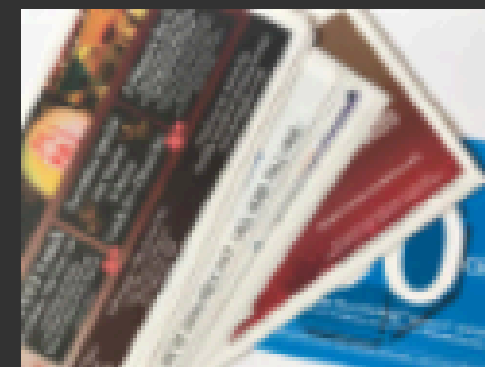
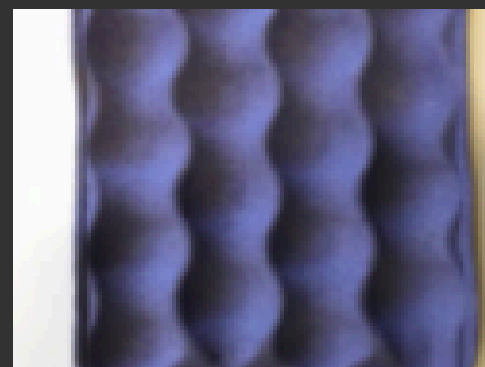
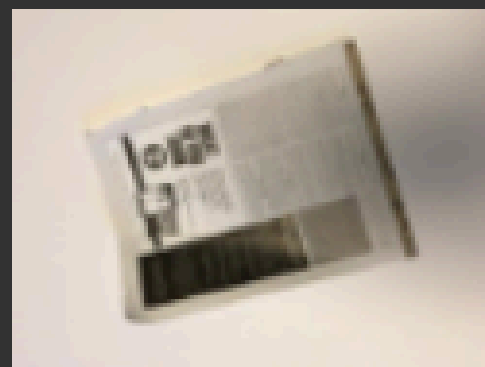
TrainingD...

glass (456)

metal (372)

paper (516)

plastic (444)





Machine Learning

2. Result by using this ML.Net Framework

Training results

Best MicroAccuracy: 0.9061
Best model: ImageClassificationMulti
Training time: 285.18 seconds
Models explored (total): 1
Generated code-behind: MLModel1.consumption.cs, MLModel1.training.cs

Best model:

MicroAccuracy: 0.9061

Model: ImageClassificationMulti

Try your model



Results

plastic 100%
glass < 1%
metal < 1%
paper < 1%

Scenario

Environment

Data

Train

Evaluate

Consume

Next steps

Evaluate

Results of training for your model can be found below.
[How do I understand my model performance?](#)

Best model:
MicroAccuracy: 0.9061
Model: ImageClassificationMulti

Try your model



Results
paper 100%
metal < 1%
plastic < 1%
glass < 1%

[Feedback](#)[Try another image](#)

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Try your model



Results
glass 99%
plastic 1%
metal < 1%
paper < 1%

[Feedback](#)[Try another image](#)

Scenario

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Results
metal 100%
glass < 1%
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paper < 1%

[Feedback](#)[Try another image](#)



Machine Learning

Best model:

MicroAccuracy: 0.9061

Model: ImageClassificationMulti

Try your model



Results

plastic 100%
glass < 1%
metal < 1%
paper < 1%

Best model:

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Try your model



Results

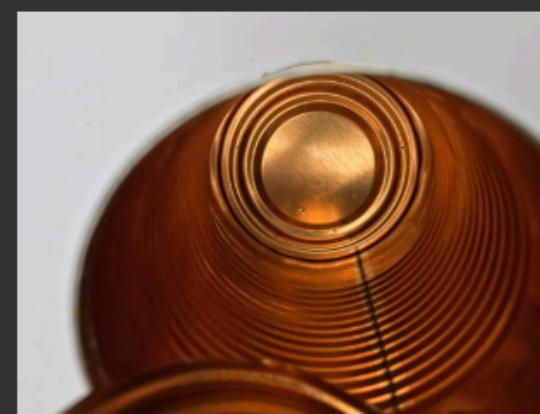
plastic 90%
glass 10%
metal < 1%
paper < 1%

Best model:

MicroAccuracy: 0.9061

Model: ImageClassificationMulti

Try your model



Results

metal 100%
plastic < 1%
glass < 1%
paper < 1%

Best model:

MicroAccuracy: 0.9061

Model: ImageClassificationMulti

Try your model



Results

metal 100%
plastic < 1%
paper < 1%
glass < 1%

Best model:

MicroAccuracy: 0.9061

Model: ImageClassificationMulti

Try your model



Results

paper 100%
plastic < 1%
metal < 1%
glass < 1%

Best model:

MicroAccuracy: 0.9061

Model: ImageClassificationMulti

Try your model



Results

paper 96%
glass 3%
metal 2%
plastic < 1%

Best model:

MicroAccuracy: 0.9061

Model: ImageClassificationMulti

Try your model



Results

plastic 94%
paper 6%
glass 1%
metal < 1%

Best model:

MicroAccuracy: 0.9061

Model: ImageClassificationMulti

Try your model



Results

glass 100%
metal < 1%
plastic < 1%
paper < 1%



COSTING

TOTAL COST:
RM 808.85

Purchased Items							
No	Items	Price for each (RM)	Qty	Total	Remark		
1	Square Hollow Iron for frame around 15	92	1	92			
2	V-Aluminium(1meterx3units), Coupling s	190	1	190			
3	CNC SHIELD ITEMS	9	1	9			
4	T8 Lead Screw diameter 8mm	25	3	75			
5	Aluminium Shaft Coupling	5.61	3	16.83			
6	KP KFL Block Ball Bearing	4.88	6	29.28			
Total		412.11					
Borrowed Items							
No	Items	Price for each (RM)	Qty	Total	Remark		
1	Arduino mega 2560	55.9	1	55.9			
2	Stepper motor Nema17	26.9	3	80.7			
3	Motor Driver DRV 8825	5.28	3	15.84			
4	UK Plug AC To DC Power Adapter 5V 3A /	13.9	1	13.9	Power supply for arduino		
5	AC TO DC 12V POWER SUPPLY 5A (60W)	19	1	19	Power Supply for 3 stepper motor		
6	Servo Motor (MG996R) 180'	9.9	1	9.9	For segregate mechanisms		
7	Touch Sensor TTP223	2.9	1	2.9	To open lid		
8	Servo Motor (SG90) 180'	6.5	1	6.5	To open lid		
9	Breadboard 400 holes	2.5	1	2.5			
10	Jumper Wire 140 pcs	6.86	1	6.86			
11	set v-aluminium-wheel(wheel(10x24mm),	23.9	3	71.7			
Total		285.7					
UnIdentified Items							
No	Items	Price for each (RM)	Qty	Total	Remark		
1	LCD Display	8.4	1	8.4			
2	Ultrasonic Sensor	3.16	4	12.64			
3	3d printed part estimation	10	1	10			
4	Transportation/Logistic/Delivery Estim	5	6	30			
5	Tools/Screw/Nut	50	1	50			
Total		111.04					
TOTAL ALL		808.85					



THANK YOU

for your time and attention